Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP300B001-V9

Revision History

DOC. Version	DATE	DESCRIPTION		CHANGED BY
00	2018-10-24	Spec	First issue	

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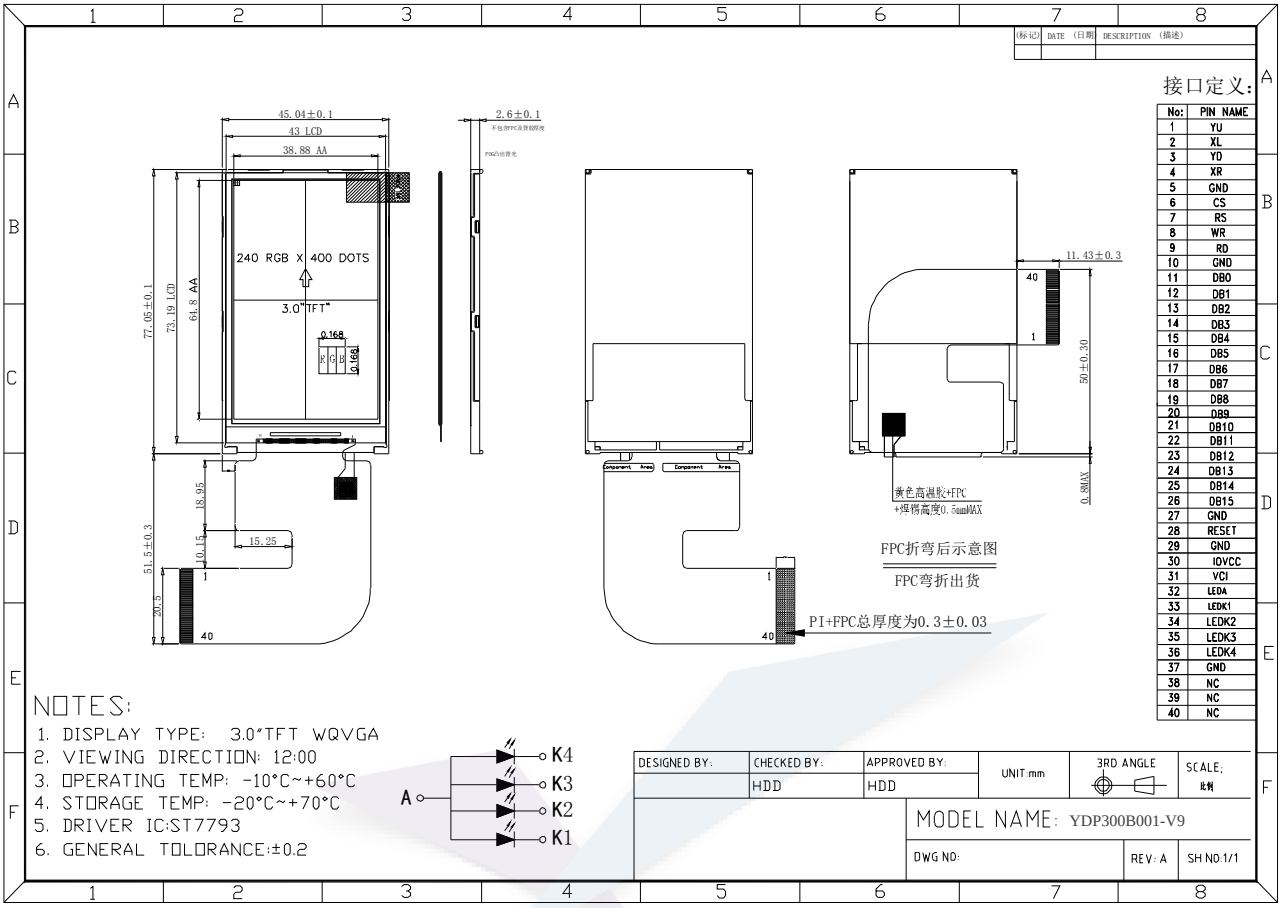
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3.GENERAL SPECIFICATIONS

ITEM	Normally dimensions
Display Type	TFT(262k) Negative Transmissive
Number of Pixel	240RGB(H)×400(V)
Module dimension	45.04(W)*77.05(H)*2.6(T)
Active Area	38.88(W)*64.8(H)
Rubbing Direction	12:00
Pixel pitch	0.168*0.168
Interface	8/16 bits parallel interface
Controller IC	ST7793
Backlight	4 CHIP White LED In Parallel

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4. LCM Outline



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5. PIN DESCRIPTION

Pin no.	Symbol	Description
1	YU	No connect
2	XL	No connect
3	YD	No connect
4	XR	No connect
5	GND	Ground
6	CS	Chip select input pin
7	RS	Data or command select signal input
8	WR	Write select signal
9	RD	Read select signal
10	GND	Ground
211-26	DB0-DB15	Data bus
27	GND	Ground
28	RESET	Reset signal
29	GND	Ground
30	IOVCC	Power supply for LCM（1.8V）
31	VCI	Power supply for the analog circuit. 2.8V
32	LEDA	Anode of Backlight
33	LEDK1	Cathode of Backlight
34	LEDK2	Cathode of Backlight
35	LEDK3	Cathode of Backlight
36	LEDK4	Cathode of Backlight
37	GND	Ground
38-40	NC	No connect

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6. Absolute Maximum Rating (Ta=25℃ VSS=0V)

ITEM	SYMBOL	MIN	MAX	UNITS
Power Supply Voltage	VDD/IOVCC	-0.3	+4.6	V
Operating Temperature	Topr	-20	70	℃
Storage Temperature	Tstg	-30	80	

Note1: Absolute maximum rating is the limit value beyond which the IC maybe broken.They do not assure operations.

Note2: Background color changes slightly depending on ambient temperature. This phenomenon is reversible.
Ta≦70℃: 75%RH max
Ta>70℃: absolute humidity must be lower than the humidity of 75%RH at 70℃

Note3: Ta at -30℃ will be <48hrs, at 80 ℃ will be <120hrs

7. ELECTRO-OPTICAL CHARACTERISTICS (Ta＝25℃)

Item	Symbol	Min	Typ.	Max	Unit	Test condition
Operating voltage	VDD/	2.5	2.8	3.3	V	-
Input voltage	V_{IL}	-0.3	-	0.2*V_{CC}	V	-
	VIH	0.8* V_{CC}	-	V_{CC}	V	-

8. Optical Characteristics:

Item		Symbol	Conditions	Specifications			Unit	Note
				Min	Typ	Max		
Transmittance		T(%)	T=0°	-	5.1	-	%	-
Contrast Ratio		C/R	At optimized viewing angle	400	500	-	-	(1) (2)
Response time		Tr	θ =0°	-	2	4	ms	(1) (3)
		Tf			6	12		
Viewing angle	Hor.	Left	CR≧10	-	45	-	deg.	-
		Right		-	45	-		
	Ver.	Top			50	-		
		Bottom			20	-		

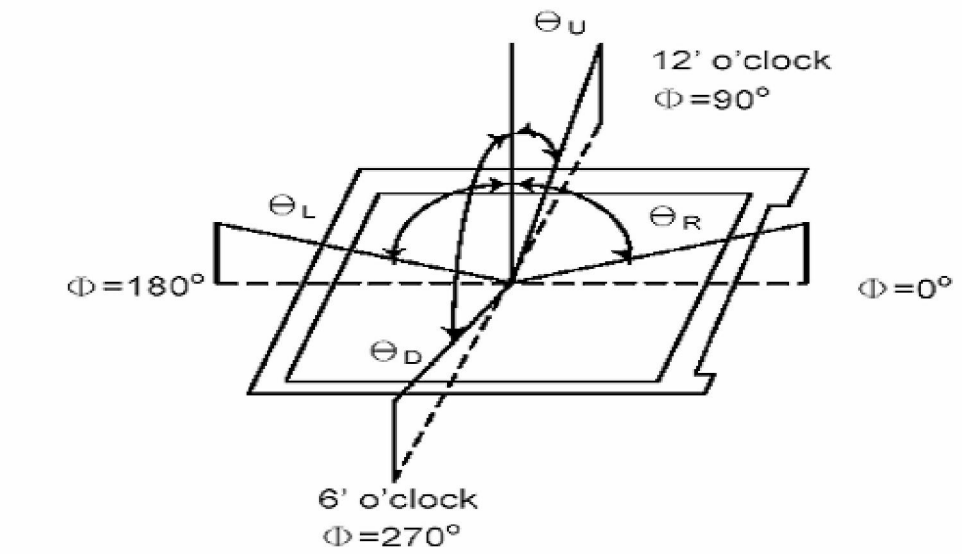
Measuring Condition
1. Measuring surrounding: dark room
2. Ambient temperature: 25±2℃
3. 30 min. Warm-up time.

Color of CIE Coordinate:

Item		Symbol	Condition	Min.	Typ.	Max.	Brightness
Chromaticity Coordinates (Transmissive)	Red	X	θ = 0°	0.606	0.626	0.646	Measured by LED light source
		Y		0.314	0.334	0.354	
	Green	X		0.257	0.277	0.297	
		Y		0.529	0.549	0.569	
	Blue	X		0.122	0.142	0.162	
		Y		0.102	0.122	0.142	
	White	X		0.283	0.303	0.323	
		Y		0.305	0.325	0.345	

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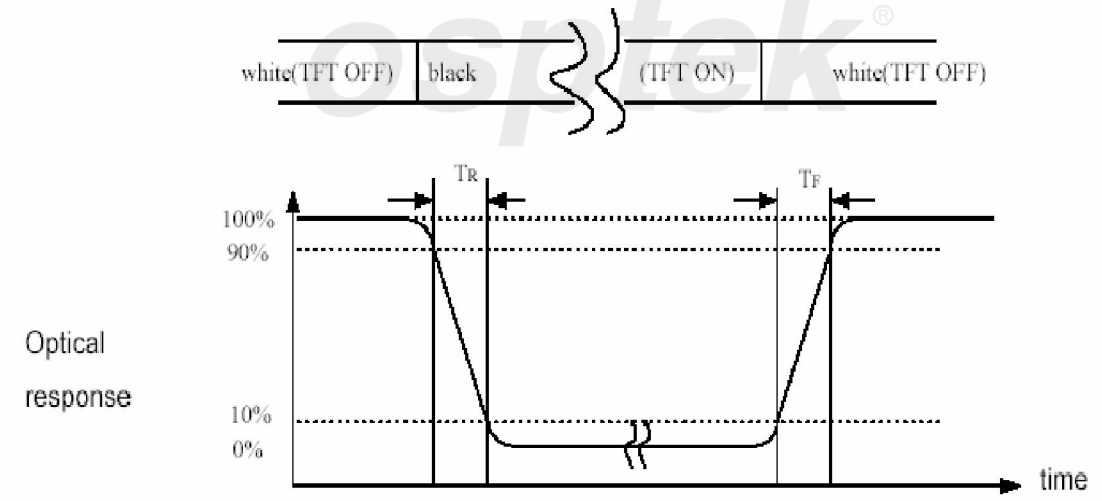
Note (1) Definition of Viewing Angle :



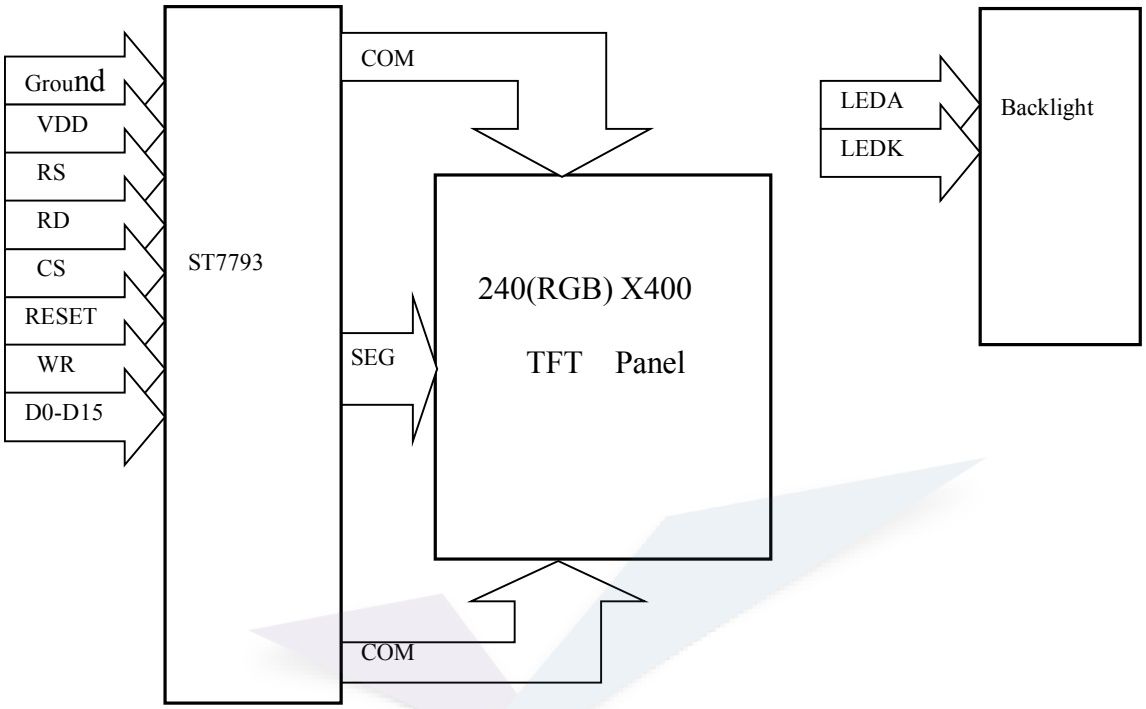
Note (2) Definition of Contrast Ratio(CR) :
measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note (3) Definition of Response Time : Sum of TR and TF



9. Block Diagram:



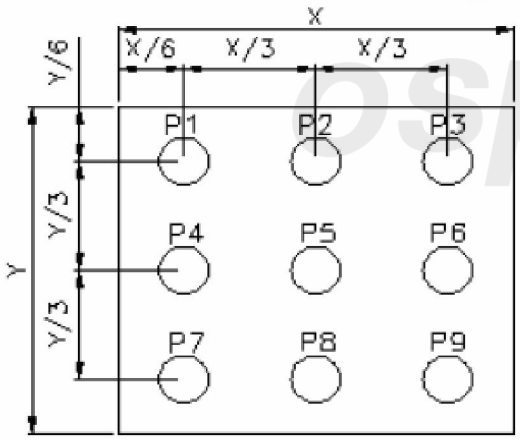
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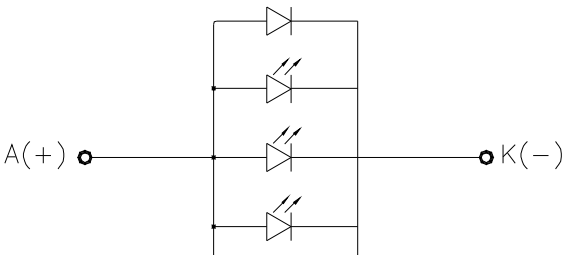
10. Backlight:

1. Standard Lamp Styles (Edge Lighting Type):
   The LED chips are distributed over the edge light area of the illumination unit, which gives the less power consumption:
2. The Main Advantages of the LED Backlight are as following:
   2.1 The brightness of the backlight can simply be adjusted.
   By a resistor or a potentiometer.
3. Data About LED Backlight:

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS	CONDITION
Froward voltage	V_F	3.0	3.2	3.4	V	I_F =80mA
Luminance	Luminance	4300	--	--	Cd/㎡	
Uniformity for LCM	Avg	80	-	-	%	
Color	**White**					

NOTE:
1. Backlight Only
2. Average Luminous Intensity of P1-P9
3. Uniformity = Min/Max * 100%
4. LED life time defined as follows: The final brightness is at 70% of original brightness
Measured Method: (X*Y: Light Area) Internal Circuit Diagram





(Effective spatial Distribution)

Hole Diameter ø3 mm; 1 to 9 per Position Measured Luminous

11. Host Interface Timing Diagrams

RGB Interface Timing Characteristics

For details, please refer to the Spec of ST7793.

12.Command table

For details, please refer to the Spec of ST7793.

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